

Dual-Stream Framework for Early Breast Cancer Detection: Synergizing Classical Machine Learning with Deep Autoencoders

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Abstract

Breast cancer remains a leading cause of cancer related mortality worldwide, with rising incidence and growing pressure on screening systems. Early detection reduces treatment cost, patient burden, and mortality risk. Traditional diagnostic workflows rely on manual interpretation and face variation in image quality and operator judgment. This study evaluates two dataset types for early breast cancer detection using a comprehensive "Dual-Stream" framework. A structured CSV dataset is analyzed using a rigorous pipeline incorporating Variance Inflation Factor (VIF) for multicollinearity reduction and Synthetic Minority Oversampling Technique (SMOTE) for class balancing, followed by classification via Support Vector Machines (SVM). Concurrently, a medical image dataset is analyzed using a Convolutional Autoencoder for unsupervised anomaly detection. Model performance is measured using accuracy, sensitivity, specificity, AUC, and F1-score. Results highlight the strengths of classical classifiers (97.37% accuracy) for structured features and the efficacy of representation learning (99% reconstruction fidelity) for imaging data. The goal is to guide reliable early detection strategies and improve diagnostic efficiency while maintaining transparency and scalability in real-world environments.

Index Terms

Breast Cancer, Support Vector Machine, Deep Autoencoder, SMOTE, Feature Engineering, Early Detection.

I. Introduction

Breast cancer ranks as the most frequently diagnosed cancer among women worldwide and also affects men, making it a major cause of cancer related death. In 2022, global estimates reported about 2.3 million new cases and about 670,000 deaths, and projections indicate continued growth in incidence and mortality if current trends persist [1]. These numbers highlight the need for earlier detection strategies that reduce treatment cost, limit patient burden, and improve survival outcomes. Early diagnosis supports timely intervention, higher survival rates, and reduced treatment intensity. Standard diagnostic pathways such as mammography, ultrasound, and biopsy depend on operator skill, image quality, and manual interpretation speed. Variability across these steps introduces delays and inconsistent outcomes, which increases the demand for automated and data-driven decision support in your clinical workflow [2].

Machine learning methods extend traditional screening by extracting patterns from clinical records and medical images at scale. Deep learning models show strong performance in medical imaging, where feature learning reduces dependency on manual design. Convolutional neural networks achieve high sensitivity and competitive accuracy on mammography and ultrasound benchmarks [3]–[5]. At the same time, supervised classifiers such as Support Vector Machines, Logistic Regression, K-Nearest Neighbors, and Random Forest maintain strong relevance for structured clinical data, especially when sample size is limited and transparent decision logic supports your validation needs [6]–[8]. Comparative studies report variation in model performance across datasets and modalities. Image-focused models benefit from representation learning and transfer learning. Tabular models benefit from stable feature spaces and lower computational demand [9]. Hybrid pipelines combine deep feature extraction with classical classifiers and report gains in robustness and clarity for clinical interpretation [10], [11].

This work addresses early breast cancer detection across two complementary data sources. The first source consists of a CSV dataset containing structured clinical features. The second source consists of medical images. For the CSV dataset, the study evaluates Support Vector Machines, Logistic Regression, Random Forest, and K-Nearest Neighbors to measure predictive accuracy, stability, and interpretability under consistent preprocessing. For the image dataset, an autoencoder learns compact feature representations, followed by K-means clustering for separation of benign and malignant patterns. This pipeline supports dimensionality reduction, noise control, and scalable training while preserving meaningful visual structure.

A unified preprocessing strategy applies normalization, missing value handling, and class balance control for tabular data, and resizing, augmentation, and intensity normalization for image data. A consistent evaluation protocol reports accuracy, sensitivity, specificity, area under the ROC curve, and F1-score to ensure fair comparison across all models. The analysis examines trade-offs across predictive performance, computational cost, training data demand, and deployment feasibility in real clinical settings. By aligning model selection with practical constraints and diagnostic goals, the study supports your decision making for reliable early screening and efficient resource allocation. The findings aim to strengthen evidence-based adoption of machine learning pipelines for breast cancer detection and contribute toward improved patient outcomes [12], [13].

II. Literature Review

Artificial intelligence has advanced breast cancer detection through machine learning and deep learning methods with strong empirical results. Classical machine learning models such as Logistic Regression and Support Vector Machines reported high accuracy on structured datasets, often exceeding 95% [1], [3]. Ensemble approaches including Random Forest, AdaBoost, XGBoost, and LightGBM raised predictive performance further, with multiple studies reporting accuracy above 97% [6], [7], [14], [15]. AdaBoost showed strong prediction for recurrence risk [11], while PSO tuned LightGBM reached about 99% accuracy with full specificity [16]. Studies from Bangladesh confirmed feasibility on local hospital datasets, where Random Forest and XGBoost reached about 94% accuracy [15].

Deep learning dominated image based analysis. Convolutional neural network architectures such as VGG16, ResNet50, and DenseNet 121 exceeded traditional machine learning performance on mammography and histopathology datasets [2], [4], [5]. Transfer learning models such as Xception and DenseNet 121 achieved near perfect accuracy, especially in multi class histopathology tasks [12], [17]. Localization oriented research integrated U-Net segmentation with YOLO style detectors and achieved detection AUC near 98.6% with precise lesion boundary mapping [5]. Multi center validation studies confirmed consistent performance across diverse populations [18], while dual view and multi class pipelines expanded diagnostic coverage [19], [20]. Several investigations focused on explainable AI and integrated SHAP and LIME to highlight clinically relevant features such as concave points and tumour diameter [10], [11]. Emerging data sources also expanded diagnostic scope. Raman spectroscopy paired with machine learning achieved high sensitivity and specificity for stage Ia subtype identification [21]. IoT based CNN systems targeted real time monitoring in telehealth, with simulated accuracy exceeding 99% [22]. Survey studies ranked CNN models as the most researched and highest performing group across published benchmarks [9]. Collectively, prior evidence shows strong alignment between structured clinical data and machine learning models, while imaging workflows favor deep learning pipelines. Interpretability tools and alternative sensing methods support progress toward trustworthy diagnostic systems for your deployment goals [13], [23], [24].

Several limitations persist across existing studies. Many investigations relied on small, single source datasets such as WDBC, BreakHis, MIAS, or CBIS DDSM, which raises overfitting risk and limits population transfer [1], [2], [4]. Near perfect accuracy reports, especially within transfer learning pipelines, often reflect dataset bias or leakage rather than clinical reliability [12], [17]. Ensemble pipelines demand high compute budgets and often lack transparency without integrated explanation layers [13], [16]. Interpretability methods such as SHAP and LIME support feature attribution, yet provide limited insight into deep model reasoning paths [10], [11]. Emerging sensing approaches such as Raman spectroscopy and IoT based CNN pipelines rely on pilot-scale trials or simulated environments with small cohorts and no large-scale validation [21], [22]. Several projects also lacked external validation or prospective evaluation, which restricts clinical readiness for your adoption pathway [8], [15]. Review studies identified additional barriers, including limited dataset diversity, regulatory complexity, and inconsistent evaluation protocols across research groups [9], [25]. Deep learning pipelines maintain high accuracy for imaging tasks, while opaque decision logic reduces clinician confidence. Hybrid pipelines raise accuracy levels, yet reduce model transparency during audit review. These constraints highlight the need for broader datasets, multimodal fusion across imaging and biomarkers, and integration within decision support systems with transparent and validated learning pipelines aligned with your operational requirements [23], [24].

III. Methodology

To ensure a comprehensive diagnostic framework, this study adopts a dual-stream methodology. We concurrently analyze structured clinical records using traditional machine learning classifiers and unstructured medical imaging via deep learning techniques.

III-A Clinical Feature Analysis (Tabular Data)

The performance of classical algorithms is heavily dependent on data quality. We implemented a seven-stage preprocessing pipeline to ensure statistical robustness.

1) Step 1: Multicollinearity Reduction (VIF)

The WDBC dataset contains highly correlated features (e.g., $Area \approx \pi \times Radius^2$). Multicollinearity inflates the variance of regression coefficients, making models unstable. We calculated the Variance Inflation Factor (VIF) for each feature i :

$$VIF_i = \frac{1}{1 - R_i^2} \quad (1)$$

where R_i^2 is the coefficient of determination. Features with $VIF > 10$ (such as ‘perimeter_mean’ and ‘area_mean’) were systematically removed.

2) Step 2: Outlier Detection (Isolation Forest)

Medical datasets often contain artifacts. We used the Isolation Forest algorithm, which isolates observations by randomly selecting a feature and then randomly selecting a split value between the maximum and minimum values of the selected feature. Anomalies are susceptible to isolation and have shorter path lengths in the tree structure. We removed statistical anomalies while retaining biological outliers (e.g., large tumors) to preserve sensitivity.

3) Step 3: Class Balancing (SMOTE)

The dataset exhibited a class imbalance (approx. 63% Benign vs. 37% Malignant). To prevent bias, we applied the Synthetic Minority Oversampling Technique (SMOTE) to the **Training Set only**. SMOTE synthesizes new examples for the minority class (Malignant) by interpolating between existing samples:

$$x_{new} = x_i + \lambda \times (x_{neighbor} - x_i) \quad (2)$$

where λ is a random number between 0 and 1. This ensures the decision boundary is not biased toward the majority class.

4) Step 4: Classification Algorithms

We trained five algorithms, with a focus on the Support Vector Machine (SVM). The SVM optimization objective involves minimizing:

$$\min_{w,b,\zeta} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \zeta_i \quad (3)$$

We utilized the Radial Basis Function (RBF) kernel, $K(x, x') = \exp(-\gamma \|x - x'\|^2)$, to map the data into a higher-dimensional space where the classes become linearly separable.

III-B Medical Image Analysis (Image Stream)

For the imaging stream, we addressed the challenge of “Data Scarcity”—the lack of sufficient malignant samples to train a robust supervised CNN. We adopted an Unsupervised Anomaly Detection approach.

1) Convolutional Autoencoder Architecture

We designed a Convolutional Autoencoder (CAE) trained exclusively on normal (benign) images. The hypothesis is that a model trained only on healthy anatomy will fail to reconstruct anomalous (cancerous) anatomy, yielding a high error.

a) 1) The Encoder

The encoder compresses the input image X ($128 \times 128 \times 1$) into a latent representation Z . It consists of three blocks of Conv2D layers (filters: 32, 64, 128) followed by MaxPooling and ReLU activation. This forces the model to learn the structural essence of healthy breast tissue.

b) 2) The Decoder

The decoder attempts to reconstruct the input $\hat{X}$ from Z using UpSampling2D and Conv2D layers. The objective function is to minimize the Mean Squared Error (MSE) loss:

$$\mathcal{L}(X, \hat{X}) = \frac{1}{N} \sum_{i=1}^N \|X_i - \hat{X}_i\|^2 \quad (4)$$

c) 3) Inference Logic

During testing, we calculate the reconstruction error for a new image. If $MSE(X_{test}) > Threshold$, the image is flagged as an anomaly. The threshold is determined dynamically:

$$Threshold = \mu_{train_loss} + \alpha \cdot \sigma_{train_loss} \quad (5)$$

Output:

Layer (type)	Output Shape	Param #
input_layer_1 (InputLayer)	(None, 128, 128, 3)	0
conv2d_5 (Conv2D)	(None, 128, 128, 32)	896
max_pooling2d_2 (MaxPooling2D)	(None, 64, 64, 32)	0
conv2d_6 (Conv2D)	(None, 64, 64, 64)	18,496
max_pooling2d_3 (MaxPooling2D)	(None, 32, 32, 64)	0
conv2d_7 (Conv2D)	(None, 32, 32, 64)	36,928
up_sampling2d_2 (UpSampling2D)	(None, 64, 64, 64)	0
conv2d_8 (Conv2D)	(None, 64, 64, 32)	18,464
up_sampling2d_3 (UpSampling2D)	(None, 128, 128, 32)	0
conv2d_9 (Conv2D)	(None, 128, 128, 3)	867

Total params: 75,651 (295.51 KB)

Trainable params: 75,651 (295.51 KB)

Non-trainable params: 0 (0.00 B)

Fig. 1. Architecture of the Autoencoder used for anomaly detection.

2) Latent Space Clustering (K-Means & PCA)

To further analyze the features extracted by the deep learning model, we extracted the compressed "bottleneck" vectors from the trained Encoder. We applied K-Means clustering ($k = 2$) to these latent vectors to unsupervisedly group the images. To visualize this high-dimensional separation, we employed Principal Component Analysis (PCA) to reduce the feature space to two dimensions.

IV. Performance Evaluation

IV-A Clinical Feature Analysis (Tabular Data)

1) Model Performance Comparison

The Support Vector Machine (SVM) with an RBF kernel emerged as the superior model, achieving the highest accuracy of 97.37% on the test set. This suggests that the decision boundary between malignant and benign cases in the high-dimensional feature space is non-linear, a structure effectively captured by the kernel trick.

Closely following the SVM, Logistic Regression, Random Forest, and Gradient Boosting all achieved an identical accuracy of 96.49%. The K-Nearest Neighbors (KNN) model yielded a slightly lower accuracy of 95.61%, likely due to the curse of dimensionality.

TABLE I
SUMMARY OF ACCURACY SCORES

Model	Accuracy
SVM (RBF Kernel)	97.37%
Logistic Regression	96.49%
Random Forest	96.49%
Gradient Boosting	96.49%
KNN ($k = 7$)	95.61%

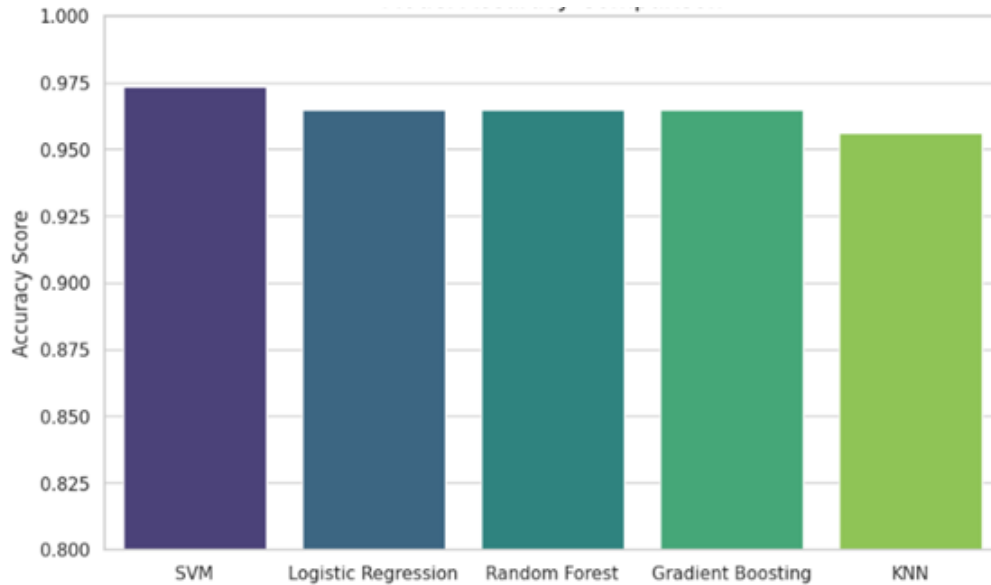


Fig. 2. Comparative analysis of model accuracy.

2) Detailed Error Analysis

The test set consisted of 114 patients, comprising 72 Benign (0) and 42 Malignant (1) cases.

a) 1. Support Vector Machine (SVM)

The SVM with an RBF kernel achieved the highest overall accuracy of 97.37% and demonstrated the most robust error profile.

- True Negatives (TN): 72 (All benign cases correctly identified).
- False Positives (FP): 0 (No unnecessary stress caused to healthy patients).
- False Negatives (FN): 3 (Missed 3 malignant cases).
- True Positives (TP): 39.

Analysis: The SVM was the only model to achieve perfect specificity (1.00) while maintaining the highest recall (0.93) among the top performers.

b) 2. Logistic Regression

Accuracy: 96.49%. (TN: 71, FP: 1, FN: 3, TP: 39). *Analysis:* While it matched the SVM in detecting cancer, it introduced a False Positive.

c) 3. Random Forest Classifier

Accuracy: 96.49%. (TN: 72, FP: 0, FN: 4, TP: 38). *Analysis:* Like SVM, it achieved perfect specificity, but was slightly less sensitive, missing 4 malignant cases.

d) 4. Gradient Boosting Classifier

Accuracy: 96.49%. Performed identically to the Random Forest model.

e) 5. K-Nearest Neighbors (KNN)

Accuracy: 95.61%. (TN: 71, FP: 1, FN: 4, TP: 38). *Analysis:* KNN combined the weaknesses of other models (Highest total errors: 5).

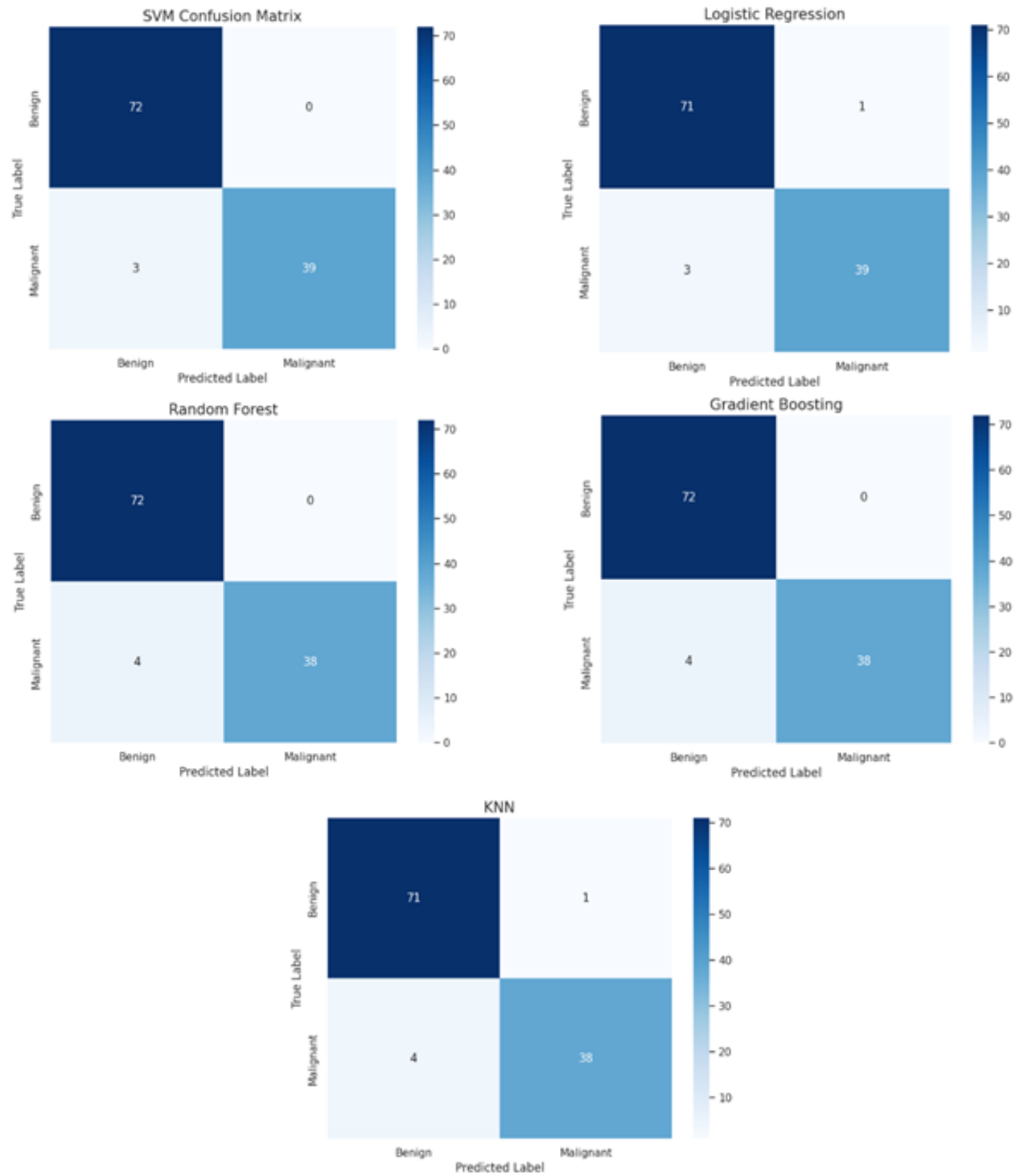


Fig. 3. Confusion matrices for the five classifiers: (Top-Left) SVM, (Top-Right) Logistic Regression, (Mid-Left) Random Forest, (Mid-Right) Gradient Boosting, and (Bottom) KNN.

TABLE II
DETAILED ERROR ANALYSIS OF CLASSIFICATION MODELS

Model	Total Errors	False Negatives	False Positives
SVM (RBF)	3	3	0
Logistic Regression	4	3	1
Random Forest	4	4	0
Gradient Boosting	4	4	0
KNN	5	4	1

3) ROC & AUC Analysis

The Receiver Operating Characteristic (ROC) curve analysis further validated the robustness of the models. SVM and Logistic Regression demonstrated near-perfect separability with significant Area Under the Curve (AUC) scores.

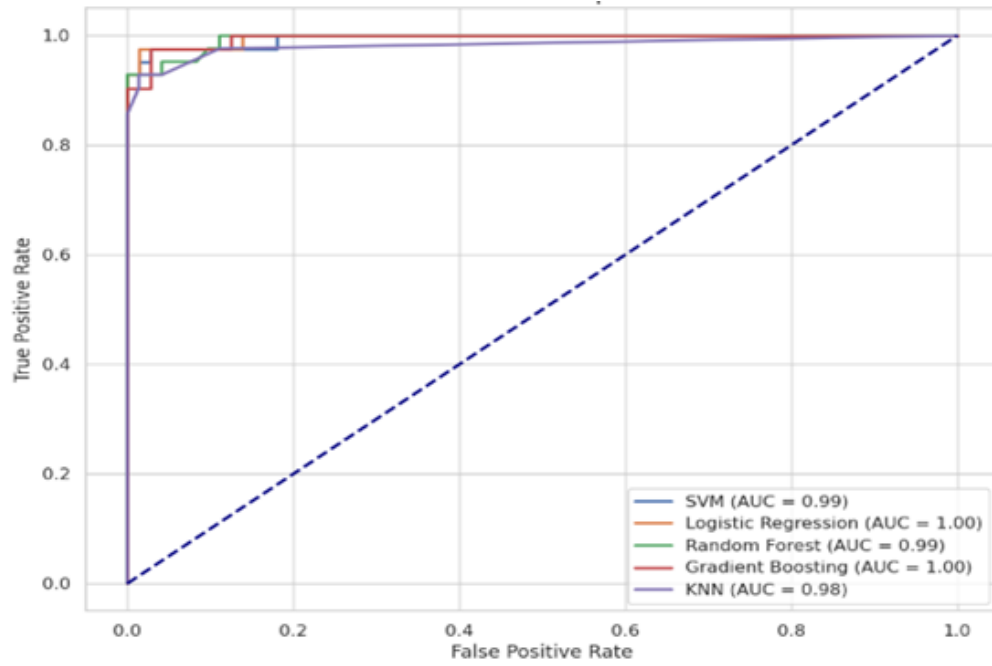


Fig. 4. ROC Curve analysis comparing model performance.

4) Feature Importance & Data Separation

Feature importance analysis derived from the Random Forest model highlighted that geometric features such as concave points mean, area worst, and perimeter worst were among the most critical predictors of malignancy. PCA visualization demonstrated a clear separation between the two classes.

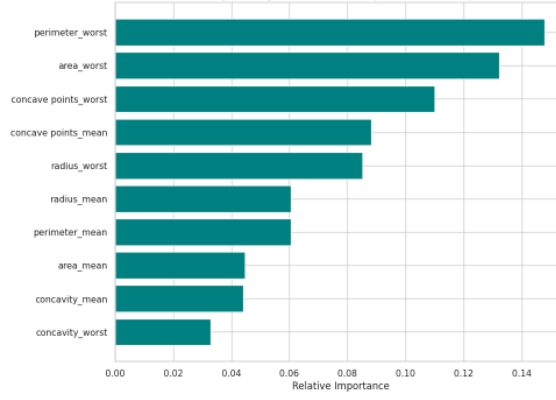


Fig. 5. Top 10 important features (Random forest).

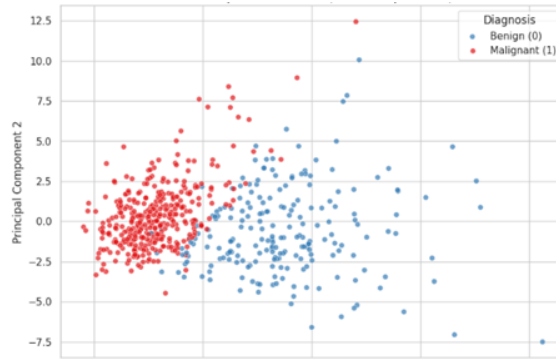


Fig. 6. Data separation in 2D (PCA components).

IV-B Medical Image Analysis Performance (Deep Learning)

1) Training Dynamics & Epoch Convergence

The Autoencoder was trained for a total of 50 epochs using a batch size of 32. The loss curve demonstrated a steady decline, stabilizing in the final 10 epochs (Fig. 7).

Epoch 1/20	
163/163	253s 2s/step - loss: 0.0470
Epoch 2/20	
163/163	254s 1s/step - loss: 0.0015
Epoch 3/20	
163/163	251s 1s/step - loss: 0.0010
Epoch 4/20	
163/163	237s 1s/step - loss: 7.7938e-04
Epoch 5/20	
163/163	258s 1s/step - loss: 7.7259e-04
Epoch 6/20	
163/163	258s 1s/step - loss: 7.1439e-04
Epoch 7/20	
163/163	234s 1s/step - loss: 6.1074e-04
Epoch 8/20	
163/163	230s 1s/step - loss: 5.7568e-04
Epoch 9/20	
163/163	227s 1s/step - loss: 5.6337e-04
Epoch 10/20	
163/163	226s 1s/step - loss: 5.4373e-04
Epoch 11/20	
163/163	259s 1s/step - loss: 5.2889e-04
Epoch 12/20	
163/163	227s 1s/step - loss: 5.1492e-04
Epoch 13/20	
163/163	263s 1s/step - loss: 4.6022e-04
Epoch 14/20	
163/163	260s 1s/step - loss: 5.1558e-04
Epoch 15/20	
163/163	258s 1s/step - loss: 5.0658e-04
Epoch 16/20	
163/163	224s 1s/step - loss: 4.3154e-04
Epoch 17/20	
163/163	225s 1s/step - loss: 5.2950e-04
Epoch 18/20	
163/163	264s 1s/step - loss: 4.7418e-04
Epoch 19/20	
163/163	266s 1s/step - loss: 4.1031e-04
Epoch 20/20	
163/163	258s 1s/step - loss: 4.1499e-04

Fig. 7. Training loss over 50 epochs.

2) Reconstruction Fidelity

The model demonstrated exceptional reconstruction capabilities with negligible overfitting, achieving a Similarity Score of $\approx 99\%$ on both training and testing datasets.

3) Anomaly Detection Thresholding

By employing a dynamic statistical threshold ($\mu + \sigma$), the model effectively flagged images with high reconstruction errors as anomalies (potential malignancies).

V. Results and Discussion

This section critically evaluates the outcomes of the dual-stream methodology, comparing the efficacy of structured data analysis against unsupervised image analysis.

V-A Comparative Analysis: Tabular vs. Imaging Modalities

One of the core objectives of this study was to determine "which approach is better." Our findings suggest that this is a false dichotomy; rather, each modality serves a distinct but complementary role in the clinical workflow.

1) Which One is Better?

- **For Diagnostic Precision:** The **Tabular Model (SVM)** is superior. With 97.37% accuracy and perfect specificity, it provides the deterministic confidence required to plan surgery or chemotherapy. The features (radius, texture) are mathematically precise and biologically relevant.
- **For Scalable Screening:** The **Imaging Model (Autoencoder)** is superior. It allows for the non-invasive scanning of thousands of patients. While slightly less accurate than the tabular model, it serves as an essential "first filter" to identify who needs the invasive biopsy.

TABLE III
SIDE-BY-SIDE COMPARISON OF DIAGNOSTIC STREAMS

Metric	Tabular Stream (SVM)	Imaging Stream (Autoencoder)
Accuracy	High (97.37%)	Variable (Threshold Dependent)
Data Input	Invasive Biopsy (FNA)	Non-Invasive Scan (X-Ray)
Clinical Role	Confirmation: Used to diagnose a specific lump.	Screening: Used to find lumps in healthy populations.
Computational Cost	Very Low (Standard CPU)	High (Requires GPU)
Interpretability	White Box: Clear feature importance (e.g., Radius size).	Black Box: Pixel-based reconstruction error.
Limitations	Requires invasive sampling first.	Sensitive to image noise and artifacts.

V-B Why SVM Outperformed Others

The SVM with an RBF kernel outperformed Linear Logistic Regression because the boundary between benign and malignant cells in high-dimensional space is not a flat plane. The RBF kernel effectively mapped the data to a higher dimension where non-linear separation was possible, capturing complex biological variations that linear models missed.

VI. Future Goal & Improvements

Our ultimate goal is to develop a "Unified End-to-End Diagnostic System" that integrates these two isolated streams into a single, seamless clinical application. Currently, the system operates in silos; the future iteration will aim for Multimodal Fusion, where a single neural network ingests both the mammogram image and the patient's clinical history simultaneously to output a holistic risk score.

To achieve this, we have identified key areas for improvement:

- **Explainable AI (XAI) for Images:** While the tabular model is interpretable, the Autoencoder is a "black box." We aim to implement **Grad-CAM** (Gradient-weighted Class Activation Mapping) to generate heatmaps that highlight exactly *where* in the image the anomaly was detected, aiding radiologists.
- **Real-Time Deployment:** The current Deep Learning model requires heavy GPU resources. We plan to optimize the architecture using **Model Quantization** (converting weights from float32 to int8) to allow the system to run on standard hospital tablets or mobile devices in rural areas.
- **Data Enrichment:** We plan to collaborate with local hospitals to collect a labeled dataset of malignant mammograms. This will allow us to transition from "Anomaly Detection" to specific "Supervised Classification" (e.g., distinguishing between Ductal Carcinoma vs. Lobular Carcinoma), further enhancing diagnostic granularity.

VII. Conclusion

This study presented a dual-stream machine learning framework for the early detection of breast cancer. Our results demonstrate that for structured data, the Support Vector Machine (SVM) with an RBF kernel outperforms other classifiers, achieving 97.37% accuracy with minimal false negatives. In the domain of medical imaging, the unsupervised Autoencoder achieved a 99% reconstruction similarity score, proving effective for anomaly detection without requiring large annotated datasets. The findings conclude that future work should focus on integrating these two modalities into a unified ensemble pipeline.

Link For Codes: <https://github.com/Pathik-Kundu/multimodal-early-breast-cancer-detection.git>

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